Assignment #7

Samuel Nasman

Problem D.9

- Using PD control, what is the system type?
- Steady state error for step, ramp, and parabola?
- How does this change with an added integrator?
- For a constant disturbance input (such as inaccurate spring constant), what is steady state error w/ and w/o integrator?

Solution:

Assuming un-modified PD control reference input error...

$$E(s) = \frac{1}{1+P(s)C(s)} R(s)$$

$$\Rightarrow P(s)C(s) = (k_D s + k_P)\left(\frac{1/m}{s^2 + \frac{b}{m}s + \frac{k}{m}}\right) = \frac{\frac{1}{m}(k_D s + k_P)}{s^2 + \frac{b}{m}s + \frac{k}{m}}$$

$\Rightarrow$ No free integrators so system is type 0 — type

Steady state error...

$$e_{ss} = \lim_{s\to 0} s \frac{1}{1+PC} R = \lim_{s\to 0}\left(s \frac{s^2 + \frac{b}{m}s + \frac{k}{m}}{s^2 + \frac{b}{m}s + \frac{k}{m} + \frac{1}{m}(k_D s + k_P)} R\right)$$

For step, $R = \frac{1}{s}$

$$e_{ss} = \frac{k/m}{\frac{k}{m} + k_P/m} = \frac{k}{k + k_P}$$ — step e_{ss}

For ramp, $R = 1/s^2$ ...

$$e_{ss} = \lim_{s\to 0}\left(\frac{k}{k+k_P} \cdot \frac{1}{s}\right) = \infty$$ — ramp e_{ss}

For parabola, $R = 1/s^3$...

$$e_{ss} = \lim_{s\to 0}\left(\frac{k}{k+k_P} \cdot \frac{1}{s^2}\right) = \infty$$ — parab e_{ss}

If we add an integrator...

$$\Rightarrow P(s)C(s) = (k_D s + k_P + k_I/s)\left(\frac{1/m}{s^2 + \frac{b}{m}s + \frac{k}{m}}\right) = \frac{\frac{1}{m}(k_D s^2 + k_P s + k_I)}{s(s^2 + \frac{b}{m}s + \frac{k}{m})}$$

$$\Rightarrow e_{ss} = \lim_{s\to 0} s \frac{1}{1+PC} R = \lim_{s\to 0}\left(s \frac{s(s^2 + \frac{b}{m}s + \frac{k}{m})}{s(s^2 + \frac{b}{m}s + \frac{k}{m}) + \frac{1}{m}(k_D s^2 + k_P s + k_I)} R\right)$$

For step, $R = \frac{1}{s}$...

$$e_{ss} = \lim_{s\to 0}\left(\frac{0}{k_I/m}\right) = 0$$ — step e_{ss}

For ramp, $R = 1/s^2$ ...

$$e_{ss} = \lim_{s\to 0}\left(\frac{k/m}{k_I/m}\right) = \frac{k}{k_I}$$ e_{ss} ramp

For parabola, $R = 1/s^3$ ...

$$e_{ss} = \lim_{s\to 0}\left(\frac{k}{k_I}\cdot\frac{1}{s}\right) = \infty$$ e_{ss} parab

For <u>disturabance input error for PD, $(R = \frac{A}{s})$</u>

$$e_{ss} = \lim_{s\to 0}\left(s\,\frac{P}{1+PC}R\right) = \lim_{s\to 0}\left(s\,\frac{1/m}{s^2+\frac{b}{m}s+\frac{k}{m}+\frac{1}{m}(k_D s+k_P)}\,\frac{A}{s}\right) = \frac{A/m}{\frac{k}{m}+\frac{k_P}{m}} = \frac{A}{k+k_P}$$ e_{ss} dist PD

For PID...

$$e_{ss} = \lim_{s\to 0}\left(s\,\frac{As/m}{s(s^2+\frac{b}{m}s+\frac{k}{m})+\frac{1}{m}(k_D s^2+k_P s+k_I)}\cdot\frac{1}{s}\right) = 0$$ e_{ss} dist PID

<u>Problem E.9</u>

(a1) For inner loop w/ PD control (unmodified), what is the system type of the inner loop?

(a2) What is steady state error when $\tilde{\theta}^d$ is a step, a ramp, and a parabola?

(a3) What is the system type w/ respect to input disturbance?

(b1) When the outer loop is PD control, what is the system type of the outer loop?

(b2) What is steady state error for z^d is a step, ramp, and parabola?

(b3) Repeat b2 if there's an integrator.

(b4) What is the system type w/ respect to an input disturbance for both PD and PID control?

<u>Solution:</u>

$$E_\theta = \frac{1}{1+PC}\tilde{\theta}_r$$

$$\Rightarrow PC = (k_{P\theta}+k_{D\theta}s)\frac{3l}{m_2 l^2 s^2 + 3m_1 z_e^2 s^2} = \frac{(k_{P\theta}+k_{D\theta}s)3l}{(m_2 l^2+3m_1 z_e^2)s^2}$$

$\Rightarrow$ System type 2 (2 free integrators). Then for a step input or ramp the error is 0, for a parabola it's finite, and for anything higher it's infinite.. (a1)

$$e_{ss} = \lim_{s\to 0}\left(s\,\frac{1}{1+PC}\cdot\frac{1}{s^3}\right) = \lim_{s\to 0}\left(\frac{1}{s}\cdot\frac{(m_2 l^2+3m_1 z_e^2)s^2}{(m_2 l^2+3m_1 z_e^2)s^2+(k_{P\theta}+k_{D\theta}s)3l}\right)$$

$$= \frac{m_2 l^2+3m_1 z_e^2}{3l k_{P\theta}}$$ e_{ss} parab (a2)

Then w/ respect to input disturbance...

$$\lim_{s\to 0}\left(\frac{P}{1+PC}\cdot\frac{1}{s^k}\right) = \lim_{s\to 0}\left(\frac{3l}{(m_2 l^2+3m_1 z_e^2)s^2+(k_{p\theta}+k_{D\theta}s)3l}\cdot\frac{1}{s^k}\right)$$

$$= \lim_{s\to 0}\left(\frac{3l}{3lk_{p\theta}}\cdot\frac{1}{s^k}\right) = \lim_{s\to 0}\left(\frac{1}{k_{p\theta}s^k}\right) = \text{constant}$$

$$\Rightarrow k=0 \Rightarrow \text{type } 0$$

(a3) type dist.